Exam 2 · Undated archive

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2. (21 points) Single choice questions. 3 points each.

2-1 Which of the following can change the *rotation speed* of a synchronous motor:

A. Altering the frequency of voltage applied to stator winding.
B. Altering the current magnitude applied to rotor winding.
C. The sequence of appearance of the three-phase windings on the stator

2-2 Which of the following can change the *rotation direction* of a synchronous motor:

A. Altering the frequency of voltage applied to stator winding.
B. Altering the current magnitude applied to rotor winding.
C. The sequence of appearance of the three-phase windings on the stator.

2-3 Which of the following can change the *magnitude of the internal generated voltage* E_A of a synchronous motor:

A. Altering the frequency of voltage applied to stator winding.
B. Altering the current magnitude applied to rotor winding.
C. The sequence of appearance of the three-phase windings on the stator.

2-4 The same synchronous motor is used in two cases. In *case one*, a three-phase 100V 50Hz AC voltage is applied to its stator winding. The angular speed of its rotor is ω_{m1}. The motor delivers 10 hp (horse power) to the load. In *case two*, a three-phase 100V 60Hz AC voltage is applied to its stator winding. The angular speed of its rotor is ω_{m2}. The motor also delivers 10 hp (horse power) to the load.

A. $\omega_{m1} > \omega_{m2}$
B. $\omega_{m1} < \omega_{m2}$
C. $\omega_{m1} = \omega_{m2}$

2. Single choice questions (continued)

2-5 If we reconnect an ordinary transformer into an auto-transformer, then the rated apparent power of the autotransformer can be higher than the rated apparent power of the original transformer.

A. True.
B. False.

2-6 A synchronous generator provides power to a RL series circuit. If we increase the angular speed of the rotor of the synchronous generator, which of the following is true?

A. The magnitude of the internal generated voltage E_A will increase.
B. The frequency of the output voltage will not change.
C. The power factor of the load will increase.

2-7 A synchronous machine has three-phase stator windings evenly placed on its stator with a sequence of a-c'-b-a'-c-b'. Now suppose we supply the windings with AC current with frequency of 60Hz and magnitude of 10A, and the current phasors in three sets of windings have the *same angle*. Which of the following can be correct about the generated stator magnetic field?

A. counterclockwise rotating magnetic field with constant magnitude,
B. clockwise rotating magnetic field with constant magnitude,
C. The magnetic field has time-varying magnitude but does not rotate.

3. (20 points)

A three-phase Y-connected synchronous generator is driven by a primer mover to produce 60Hz AC voltage. It has a synchronous reactance of j2 Ω per phase and negligible synchronous resistance. When the excitation current is 10A, and no load is connected to stator windings, the line-to-line voltage at the output is measured to be 100V. Now we increase the excitation current to 20A and connect a three phase Y-connected load with an impedance of 10+j10 Ω per phase to the generator. We adjust the mechanical input to the generator such that the speed of its rotor is the same as before. Under the new condition,

(10 points) Draw the single-phase equivalent circuit of the system and label the impedances.

(10 points) Find the RMS value of the line-to-line voltage at the load side.

4. (22 points)

A three-phase 1.732-kV 2-pole 60-Hz Y-connected synchronous generator has a synchronous reactance of j1Ω per phase and negligible armature resistance. The synchronous generator is driven by a turbine.

Suppose under certain load condition, the RMS value of the line-to-line terminal voltage of the generator is 1.732kV, and the generator supplies 1.5MW real power to the load. The power angle is 30° at this operating condition.

a) (7 points) How much torque is supplied by the turbine to the generator, assuming that all losses are neglected?

b) (7 points) What is the RMS value of the internal generated voltage in each phase of the generator?

c) (8 points) What is the power factor of the load?

5. (32 points)

A 600V three-phase Δ-connected synchronous motor is connected to a 600V bus. The motor has a synchronous reactance of j4 Ω per phase and negligible armature.

a) (16 points) Under certain load condition, the internal voltage lags the terminal voltage by 60° in each phase, and the developed power in the motor is 40 kW. Find the RMS value of the line current.

b) (10 points) We increase the field current by 20% and adjust the prime mover such that the motor achieves it maximum developed power. Find the new torque angle and the RMS value of the new line current.

c) (6 points) What is the reactive power the motor consumes? What is the RMS value of the current in each phase of its stator windings?